

Husein Sidharta Muhammad

AI/ML ENGINEER

Dusun Besalen, RT. 06, No. Rumah 47, Kel. Baturetno, Kec. Banguntapan, Kab. Bantul, DI Yogyakarta 55197
huseinmuhammad@student.ub.ac.id • +62 819 0432 7982

EDUCATION

Brawijaya University <i>Bachelor of Information Technology</i> Awardee of Education Scholarship of Indonesia (BPI) 2024 (August 2024 – Ongoing)	<i>Malang, Indonesia</i> August 2024 – Ongoing
National University of Singapore – Summer Program <i>Two-week hands-on course: AI for Community Impact</i>	<i>Singapore</i> 19 June 2023 – 1 July 2023
Islamic State Senior High School (MAN) 2 Yogyakarta <i>The Most Outstanding in the Class – Academic Category</i> Awardee of Indonesia Advanced Country Scholarship (BIM) Study Abroad Preparation Program • GPA: 91.67/100 • SAT: 1220	<i>Yogyakarta, Indonesia</i> June 2021 – July 2024

LEADERSHIP AND PROJECTS

SIGNESIA — BISINDO (Indonesian Sign Language) Cultural Preservation App

Malang, Indonesia • Jul 2026 – Ongoing
Brawijaya University — NLP Engineer (Text-to-Gloss)
Developing the NLP pipeline for SIGNESIA, a mobile app that preserves Indonesian Sign Language (BISINDO) by scanning QR artifacts, selecting a regional dialect (Malang, Surabaya, or Jakarta), and playing narration with per-word BISINDO skeleton animation. Designated to develop an Indonesian-to-gloss (Text-to-Gloss) translator.

BRONE — Local Indonesian Brawijaya University Voice Assistant Robot

Malang, Indonesia • Aug 2026 – Ongoing
Brawijaya University — Speech-to-Text (ASR) & LLM Engineer
Built the speech-to-text and LLM subsystems for BRONE, a fully local Indonesian voice assistant (STT → LLM → TTS) targeting under 2.5-second time-to-first-speech. For ASR, benchmarked Faster Whisper tiny/base/small on the Intel NUC and cut live word-error-rate from 17.1% to 7.2% via real-time noise suppression (RNNoise) and domain-aware handling of BRONE, FILKOM, and Brawijaya terms. For the LLM, ran local-versus-cloud experiments (Ollama Qwen 0.8B–4B on the NUC against Deepseek V4 Flash), we found that local model in NUC is not recommended while Deepseek perform well in accuracy and latency.

Collaborative Research — Calibrated Safety Layer for Preventing Expert Mis-Dispatch in Radiology Vision-Language Model VILA-M3

Malang, Indonesia • Aug 2026 – Ongoing
Intelligent System Laboratory & Kalbe Digital Lab — Member
Co-authored a GEMASTIK XIX scientific paper and qualify to the final, proposing a calibrated safety layer (a confidence-gated router) built on BioMedCLIP + MLP that prevents expert mis-dispatch in the radiology vision-language model VILA-M3. The router reduced the baseline expert mis-dispatch rate from 8% to 0% on 200 evaluation samples (McNemar’s test, $p \ll 0.001$), outperformed the best parser strategy by +17.5 percentage points, and achieved routing consistency of 1.000 across 28,600 image–query pairs.

Intelligent System Laboratory – Computer Science Faculty, Brawijaya University

Malang, Indonesia • Aug 2025 – Ongoing
Operasional Division
To actively do research and collaboration in newest AI topics with other lab members as well as laboratory lecturers. By joining the lab, I get access to resources such as NVIDIA’s high-end GPUs. As one of the members of the operasional division, besides doing research, I also help the laboratory document the lab’s events and post them to social media.

English Study Students Forum (FORMASI), Brawijaya University

Malang, Indonesia
Vice Coordinator Narrative Arts Division
September 2025 – Ongoing
FORMASI is a student activity unit providing opportunities Brawijaya University Students to boost their English skills. I actively contribute to the regular practice by preparing power point materials and reaching out a coach for the practice.

Unit Aktivitas Bulu Tangkis (UABT), Brawijaya University

Public Relation Staff

Malang, Indonesia

September 2025 – Ongoing

UABT is a student activity unit facilitates Brawijaya University Students to play badminton regularly. I contribute to the division by proposing external media partnership. I am also the person in charge for UABT anniversary, arranging event rundown, designating place of the event, and developing the staff executor structure.

SKILLS & INTERESTS

- **Technical:** LLM engineering, Python programming basic and advanced (Object-Oriented Programming), Data Science and Machine Learning, Computer Vision, Full Stack Web Development.
- **Language:** B2 in English (IELTS 6), native proficiency in Bahasa Indonesia.
- **Laboratory:** Prominent skills and experience in varied research methods.
- **Interests:** Artificial Intelligence and scientific research projects.

AWARDS AND ACHIEVEMENTS

- Finalist of Scientific Paper GEMASTIK 2026 (Topic in Visual Language Model Radiology)
- 3rd Place of National IEEE Fest UB Research Plan PKM-KC 2025 (Topic in Machine Learning)
- Favorite Champion of Student Creativity Program Writing Competition (LP2PKM) Brawijaya University 2025 – Creative Initiative (KC) Category at the University Level
- Talented Student of Student Creativity Program Writing Competition (LP2PKM) Brawijaya University 2025 – Creative Initiative (KC) Category at the University Level
- The Best Intern Staff at University Students' Research Group (K-RISMA) Brawijaya University Internship Program 2024
- 3rd Place of English Debate Competition of FORCE 2024 at the University Level
- 1st Place of Dekan Cup Computer Science Faculty Brawijaya University 2024 (Research Competition) – Information Technology Field at the Faculty Level
- Maximum Predicate (A-Distinction) in the Summer Program "AI for Community Impact" at the National University of Singapore 2023
- 1st Place in National Young Scientific Research Competition "Digital Literacy Strategy to Prevent Cybercrime and Hoax" 2022 by Surabaya State University
- Finalist of the National Level Young Researcher Competition (LPB) in the Field of Economics 2022 by the Center for Young Scientists
- 2nd Place in the Province Level Young Researcher Competition (LPB) in the Field of Economics 2022 by the Center for Young Scientists
- Top 30 National Level Scientific Research Competition "Young Generation Innovation in Enhancing Indonesia's Economics" 2022 by Management Study Program UIN Sunan Ampel Surabaya

CERTIFICATION

Dasar Linux Untuk Cyber Security

Cyberkarta · 2026

Python for Data Science and Machine Learning Bootcamp

Udemy · 2025

NLP - Natural Language Processing with Python

Udemy · 2025